

SAS Club 2024

Der Business Analytics Club für SAS User

14. November 2024

03/2000 bis 11/2024: 24,5 Jahre SAS Club 😊



Speakers



Gerhard Svolba
Data Scientist und Analytic
Solution Architect, SAS



Jannic Horst
Data Science Advisor, SAS



David Weik
Sr. Software Development
Engineer, SAS R&D



Bernhard Lainer
SAS Application Manager,
Raiffeisen Informatik



Phillipp Gnan
Data Scientist, Predictive
Analytics Competence
Center, Bundesministerium
für Finanzen



Gernot Engel
Customer Success Manager
Austria, SAS



Kashif Din
Principal Technical
Architect, SAS

Agenda

15:00	Begrüßung	Gerhard Svolba
15:10	Synthetic Data und SAS Data Maker	Gerhard Svolba
15:35	Autoencoders: die Anatomie einer Steuererklärung	Phillipp Gnan
16:00	Mit SAS & LLMs einen Use Case umsetzen und produktiv setzen	David Weik
16:25	< Pause >	
16:45	Was kann die neue Viya Workbench?	Jannic Horst
17:10	SAS Viya auf Kubernetes: Erfahrungsbericht nach zwei Jahren in Produktion	Bernhard Lainer
17:35	SAS Content Assessment - Anwendungsfälle&Nutzen - SAS support news	Gernot Engel
17:55	25 years SAS Enterprise Guide: the story continues on Viya	Kashif Din

Synthetic Data und SAS Data Maker

Sammelst du noch oder generierst Du schon??



Gerhard Svolba
Data Scientist
SAS Austria



Auf welchem Bild ist eine reale Person zu sehen?



Auf welchem Bild ist eine reale Person zu sehen?



auf keinem

Inhalt

- 1 Welche Anwendungsfälle gibt es, synthetische Daten für die Datenanalyse zu verwenden?
- 2 Mit welchen Methoden können synthetische Daten generiert werden?
- 3 Wie sieht der Prozess der Generierung und Validierung der synthetischen Daten aus

**Welche Anwendungsfälle
gibt es, synthetische Daten
für die Datenanalyse zu
verwenden?**

Warum brauchen wir synthetische Daten?

Wir haben
keinen Zugriff



No Internet Connection

Wir dürfen die Daten
nicht verwenden

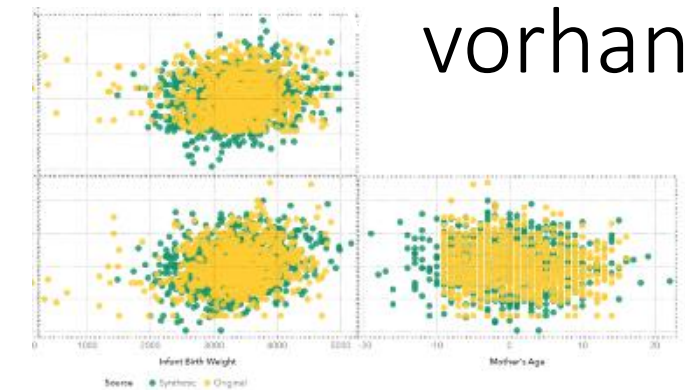


Es gibt die Daten
(noch) nicht

404
Not Found

The resource requested could not be found on this server!

Nicht in ausreichender
Menge/Qualität
vorhanden

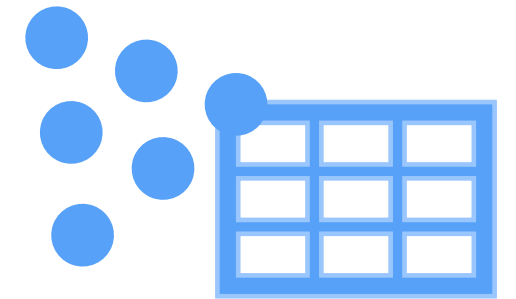


Konsequenz: lange Vorlaufzeiten und
Projekte können nicht gestartet werden

What is synthetic data?

Synthetic data is generated by applying a sampling technique to real-world data or by creating simulation scenarios where models and processes interact to create completely new data not directly taken from the real world.

– *Gartner*



Synthetic data and generative AI



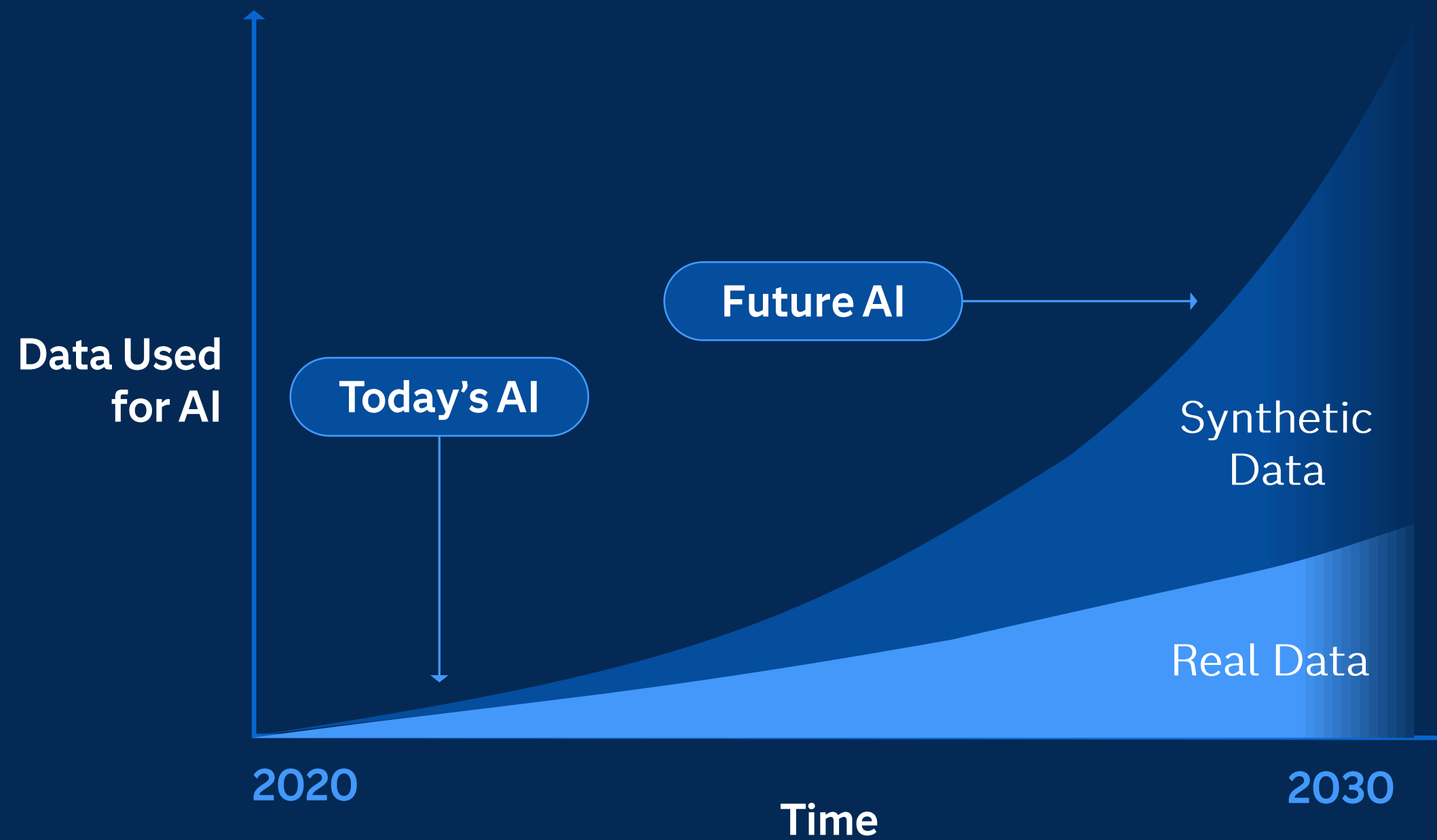
Synthetic data has become important with the emergence of GenAI given GenAI models require large amounts of data for training.



Synthetic data generation can be considered a subset of GenAI given it generates data.

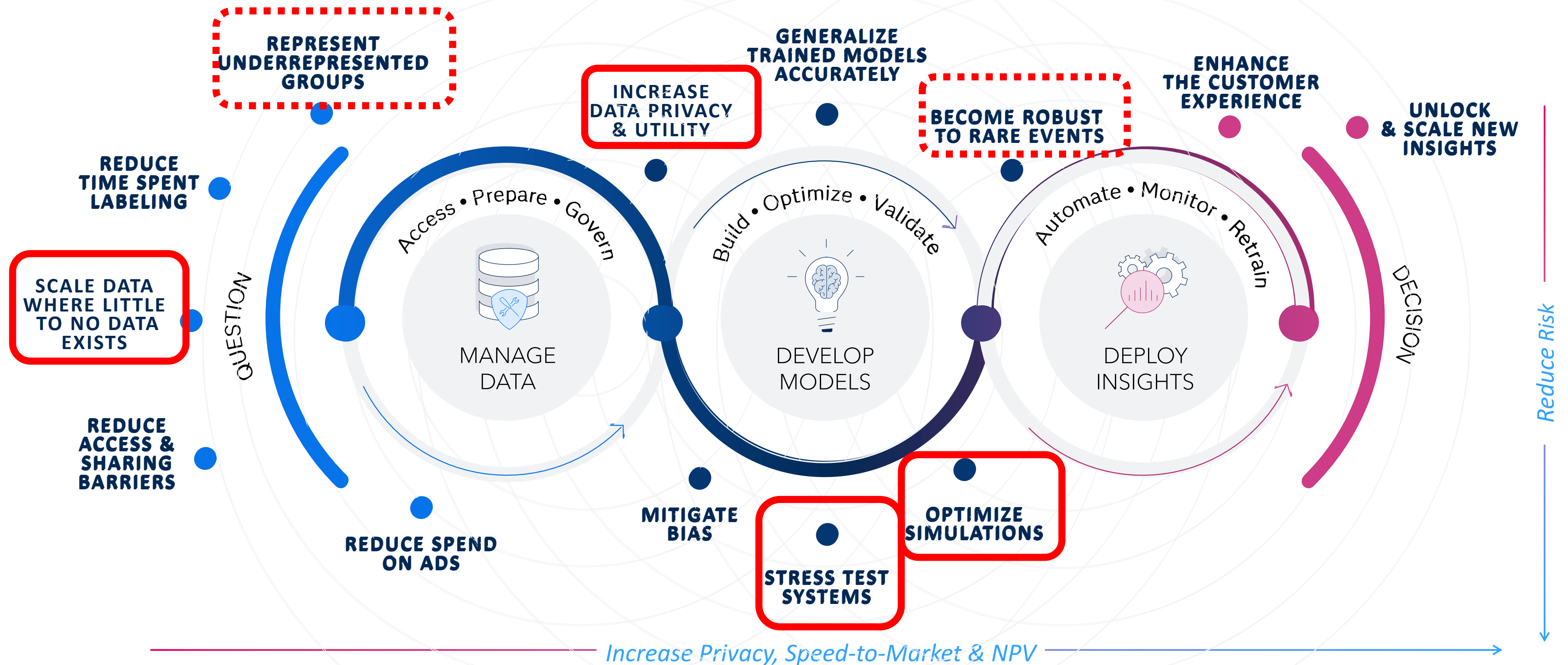


Synthetic data is also considered to be a privacy preservation enabler.



**BY 2030, SYNTHETIC
DATA WILL
OVERSHADOW REAL
DATA IN AI MODELS**

Synthetic data generation applications across the E2E AI & Analytics Lifecycle



Industry

Banking



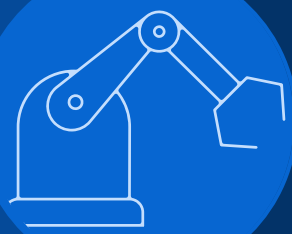
Public
Sector



Life
Sciences



Manufacturing



Use Cases

Fraud Detection

- Rare events
- Simulate fraudulent transactions
- Improve detection

Policy Development

- Simulate demographic information
- Support public policies development

Clinical Trials

- Simulate patient cohorts, treatment outcomes
- Execute clinical trials and research studies

Predictive Maintenance

- Simulate sensor data from manufacturing plants
- Develop predictive algorithms

Climate Risk

- Simulate climate related events
- Mitigate risks associated with climate change

Law Enforcement

- Simulate criminal behaviours
- Assess impact of crime reduction initiatives on public safety

Information Exchange

- Emulate electronic health records
- Facilitate health information exchange initiatives

Worker Safety

- Simulate hazardous scenarios
- Develop protocols to reduce workplace incidents

AI Applications

Data
Acquisition

Data Sharing

Bias and
Privacy

Model Training

Data
Retention

**Mit welchen Methoden
können synthetische Daten
generiert werden?**

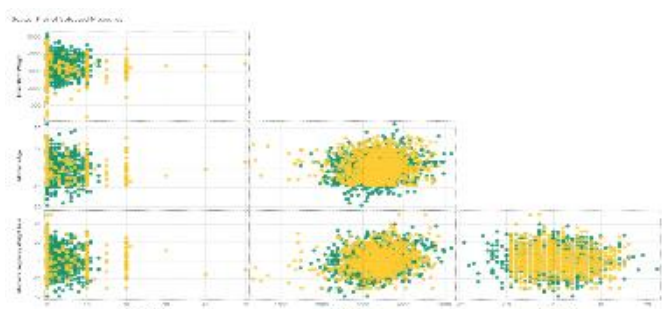
Generierung synthetischer Daten (aus der Grundgesamtheit)

Original Data	SynthPatnr	Infant Birth Weight	Black Mother	Married Mother	Mother's Education Level
	-	2693	0	1	3
	-	3033	0	1	3
	-	3147	0	1	3
	-	2807	0	1	3
	-	3856	0	1	3
	-	3090	0	1	3
	-	4309	0	1	3
	-	2807	0	1	3
	-	3750	1	1	3
	-	3600	0	1	3
	-	4196	0	1	3
	-	7035	0	1	3

Frequ...
63

Original Data

Learns



Analytical Model
+ Business Rules

Creates
(Sample)

Synthetic Data	SynthPatnr	Infant Birth Weight	Black Mother	Married Mother	Mother's Education Level
	1730	4203	0	1	3
	1014	3556	0	1	3
	1828	2656	0	1	3
	1665	2555	0	1	3
	1164	3098	0	1	3
	1445	3274	0	1	3
	1237	4067	0	1	3
	1862	4054	0	1	3
	1343	3235	0	1	3
	1107	3423	0	1	3

Frequ...
68

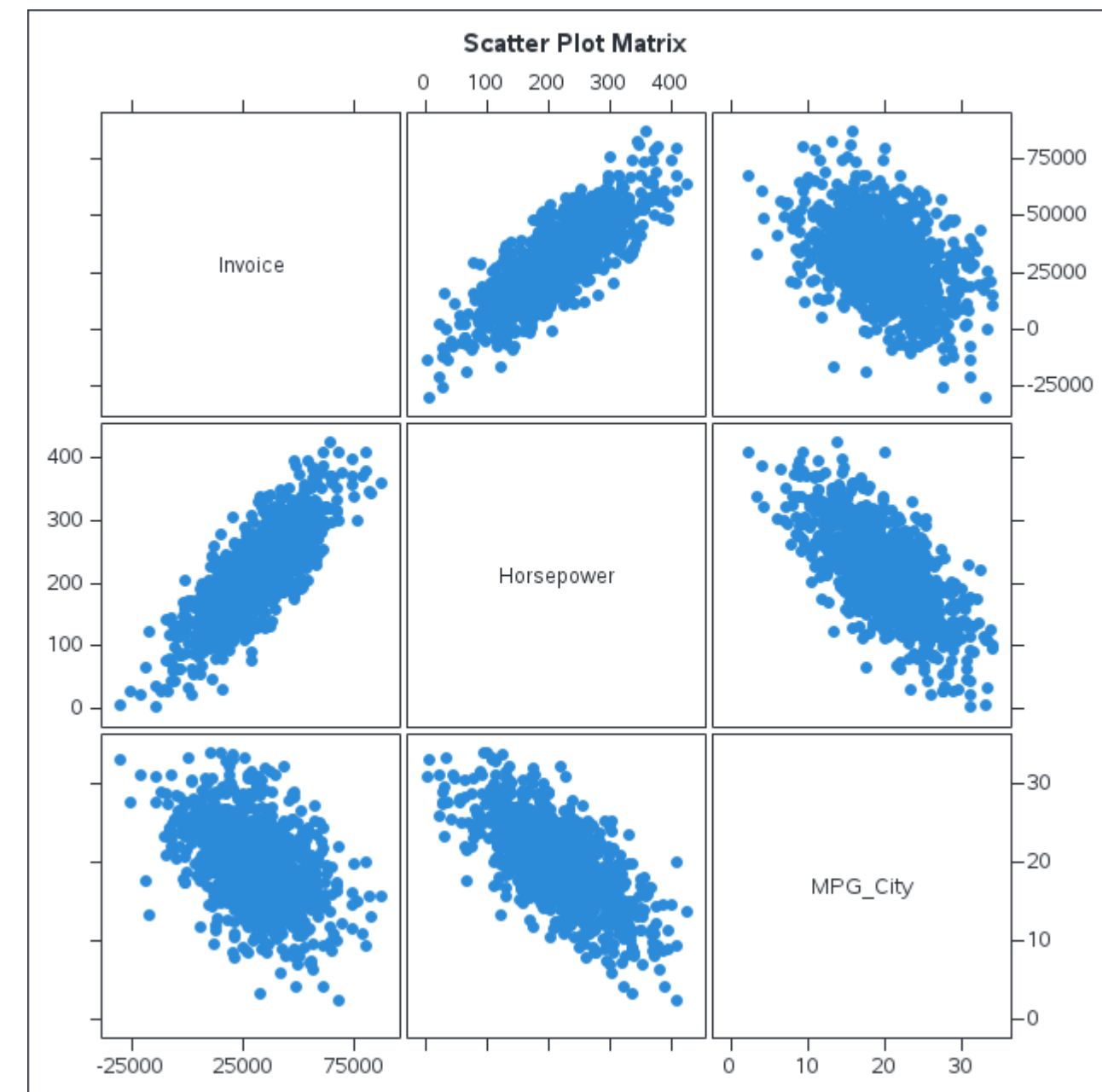
Synthetic Data

Ein einfaches Beispiel mit SAS Visual Statistics (SAS/STAT)

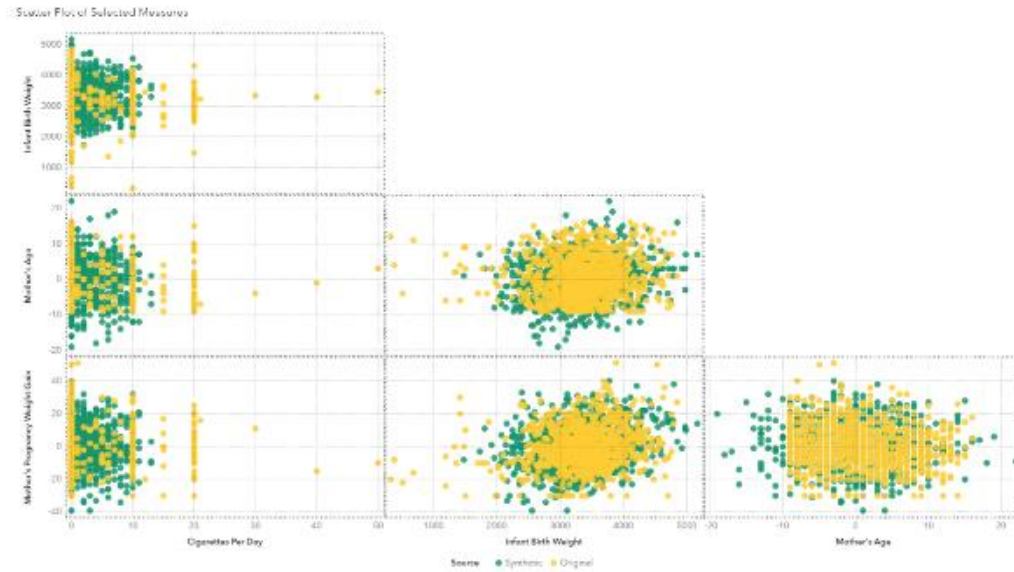
```
proc corr data=sashelp.cars  
    out=work.cars_cov cov  
    plots(maxpoints=10000000)=matrix;  
var invoice horsepower mpg_city ;  
run;
```

```
proc simnormal data=work.cars_cov(type=cov)  
    out = work.cars_synth  
    numreal= 1000  
    seed = 1;  
var invoice horsepower mpg_city ;  
run;
```

```
proc corr data=work.cars_synth cov  
    plots(maxpoints=10000000)=matrix;  
var invoice horsepower mpg_city ;  
run;
```

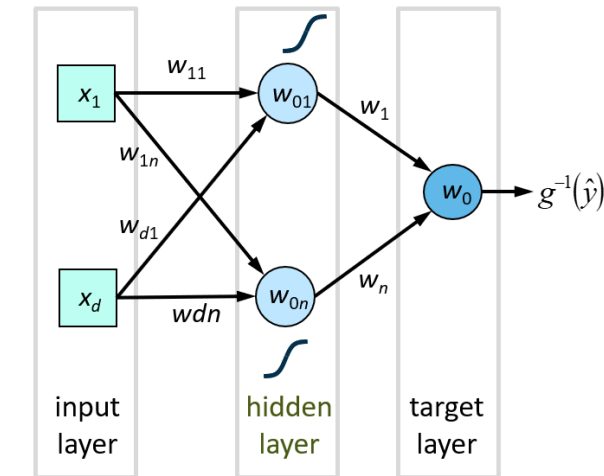
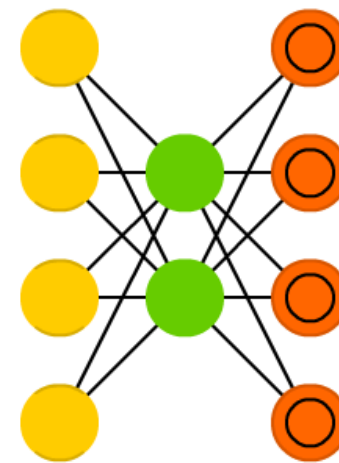


Gängige Methoden zur Datensynthetisierung



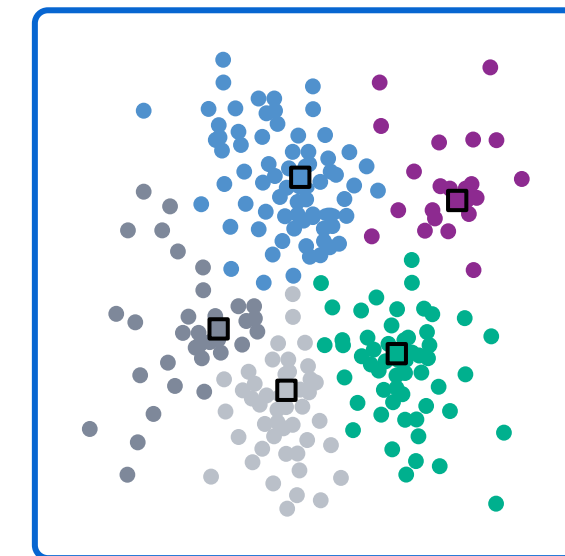
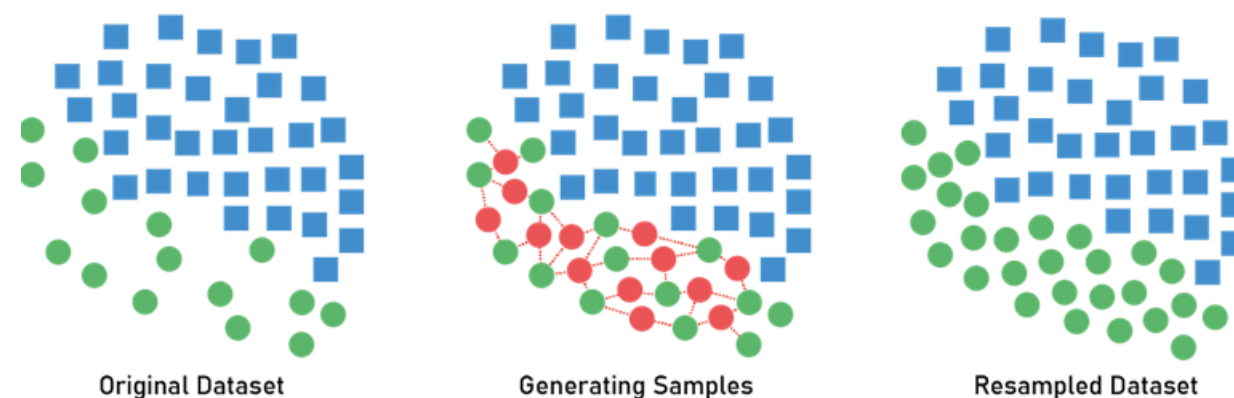
Multivariate Distributions

Auto Encoders



Generative Adversarial Networks

SMOTE - Synthetic Minority Oversampling Technique

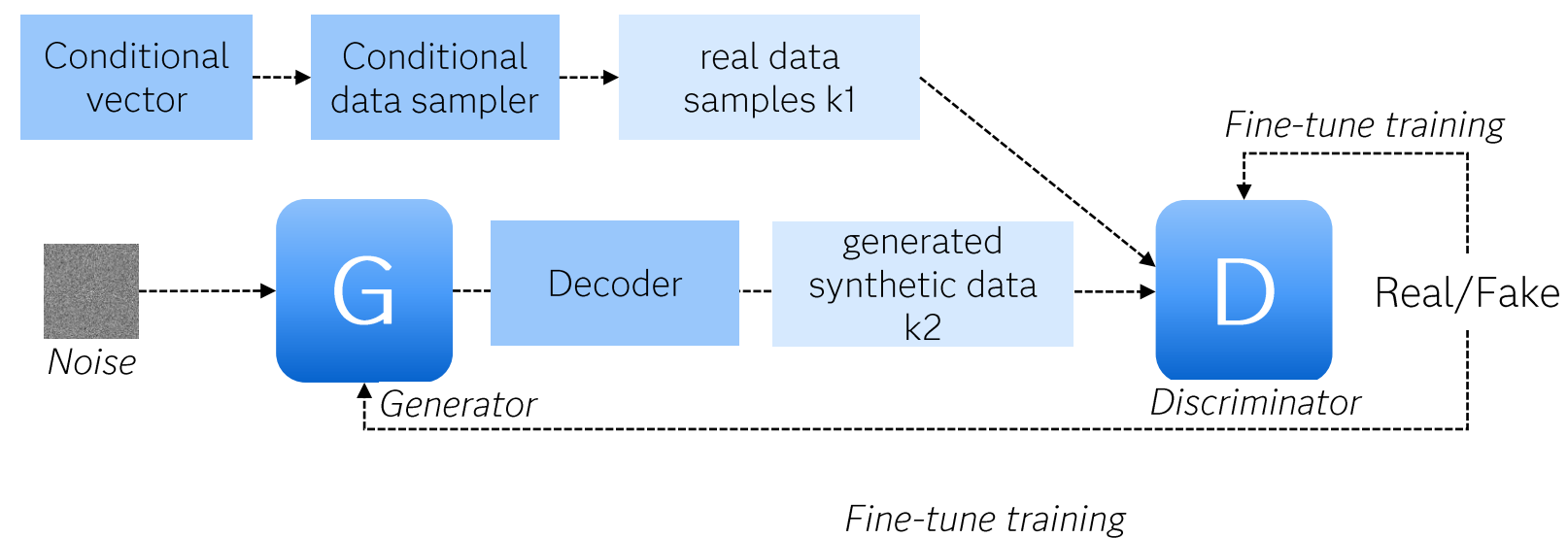


kNN k-nearest-neighbour

Common algorithms to generate synthetic data

GANs is the most used method for generating synthetic data

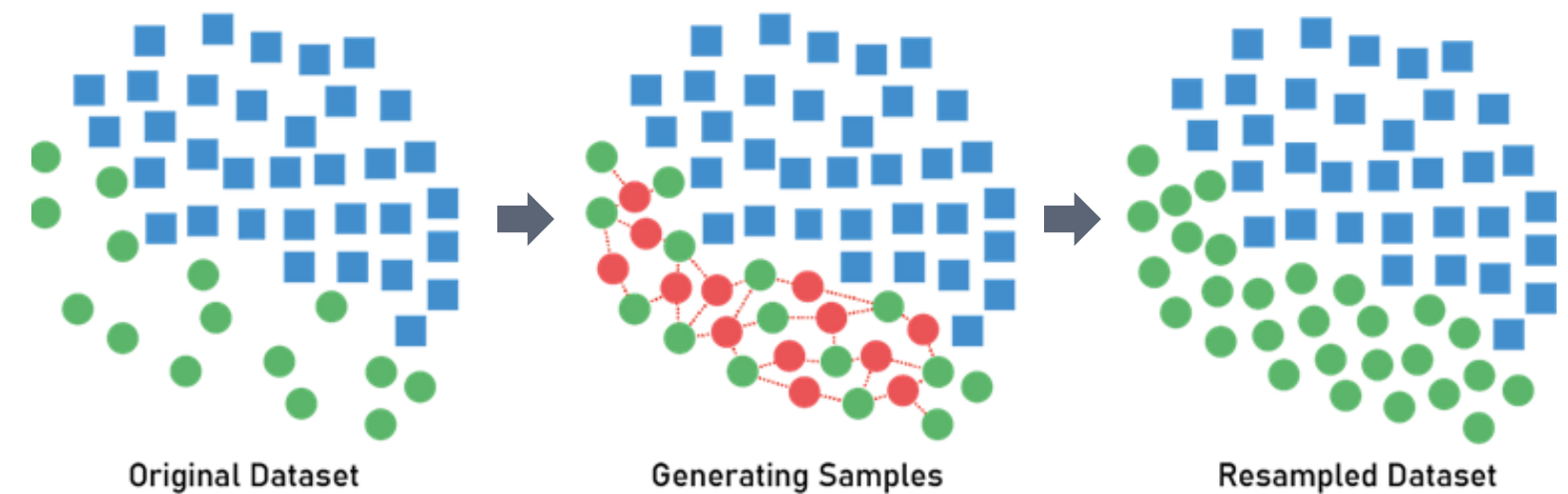
GAN – Generative Adversarial Network (CPCTGAN, tabularGAN)



- GANs are deep learning based generative models.
- It is a powerful machine learning technique which can learn to mimic a given distribution of data.
- GANs are used for generating synthetic data that is indistinguishable from real data.
- GAN Consists of a “generator” and a “discriminator” component
- It uses cluster centroids from encoded input data to learn the distribution
- Generates new data which is then assessed by the discriminator for the “probability of being fake”
- SAS provides a correlation-preserving conditional tabular GAN (CPCTGAN) model for generating synthetic tabular data.

[SAS Help Center: Generative Adversarial Network Action Set](#)

SMOTE - Synthetic Minority Oversampling Technique

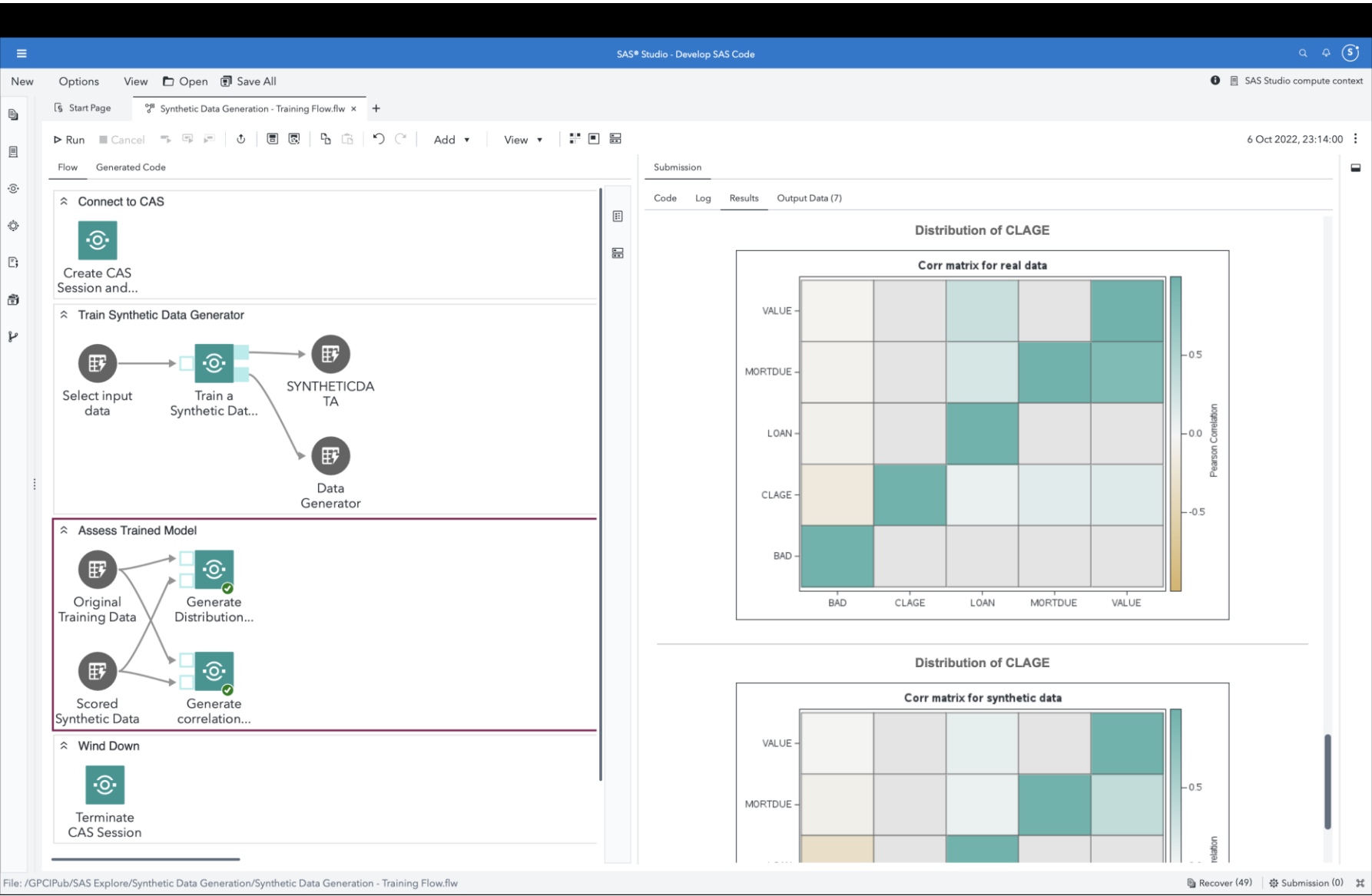


- A powerful tool that can be used to create new data and augment existing datasets.
- This data augmentation technique can be used to improve the accuracy of ML models on imbalanced datasets.
- SMOTE generates new examples from the minority class
- SAS provides smoteSample Action which uses SMOTE technique

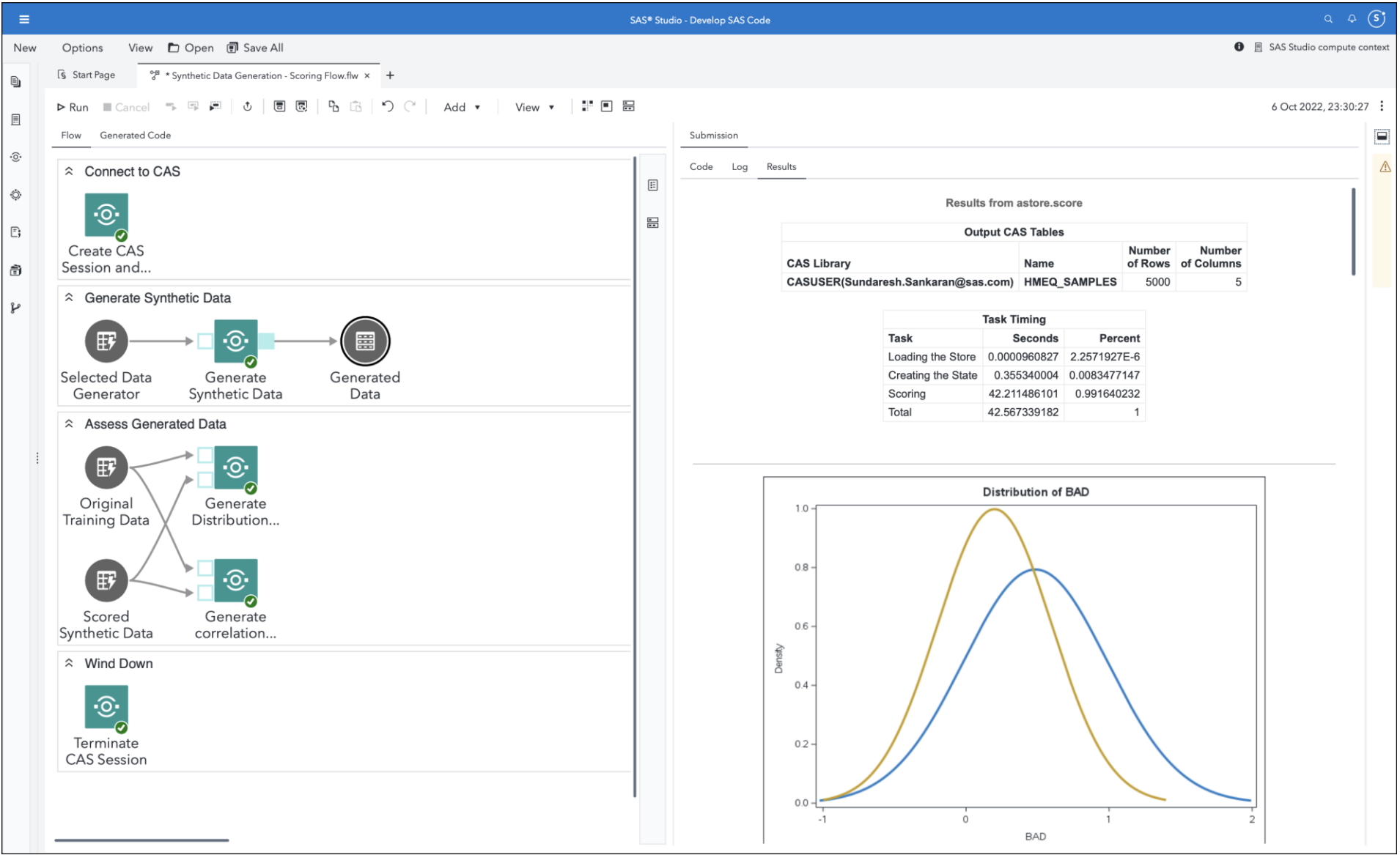
[SAS Help Center: Synthetic Minority Oversampling Action Set](#)

SAS Studio Custom Steps on SAS Viya to train, score and assess synthetic data models

Training a Synthetic Data Model



Generating Synthetic Data



[sas-studio-custom-steps/Synthetic Data Generation at main · sassoftware/sas-studio-custom-steps · GitHub](https://github.com/sasstudio-custom-steps/Synthetic-Data-Generation)

Example: SAS Data Maker

SAS Data Maker Capabilities

Address gaps and limitations in real-world data

1

Generate data on demand using algorithms. Enable pre- and post-processing data



2

Replicate data characteristics including statistical properties, distributions



3

Evaluate synthetic data quality with visual, evaluation metrics



SAS® Data Maker

Your gateway to seamless synthetic data generation

Select preprocessed tabular source data:

Select Data

Search area: ⓘ

3

Number of records to generate: ⓘ

10000

Generate Data



How To



Overview

See an overview of SAS Data Maker.

[Watch video](#)



Get Started

Learn about Data Maker capabilities and how to get started with SAS.

[View documentation](#)

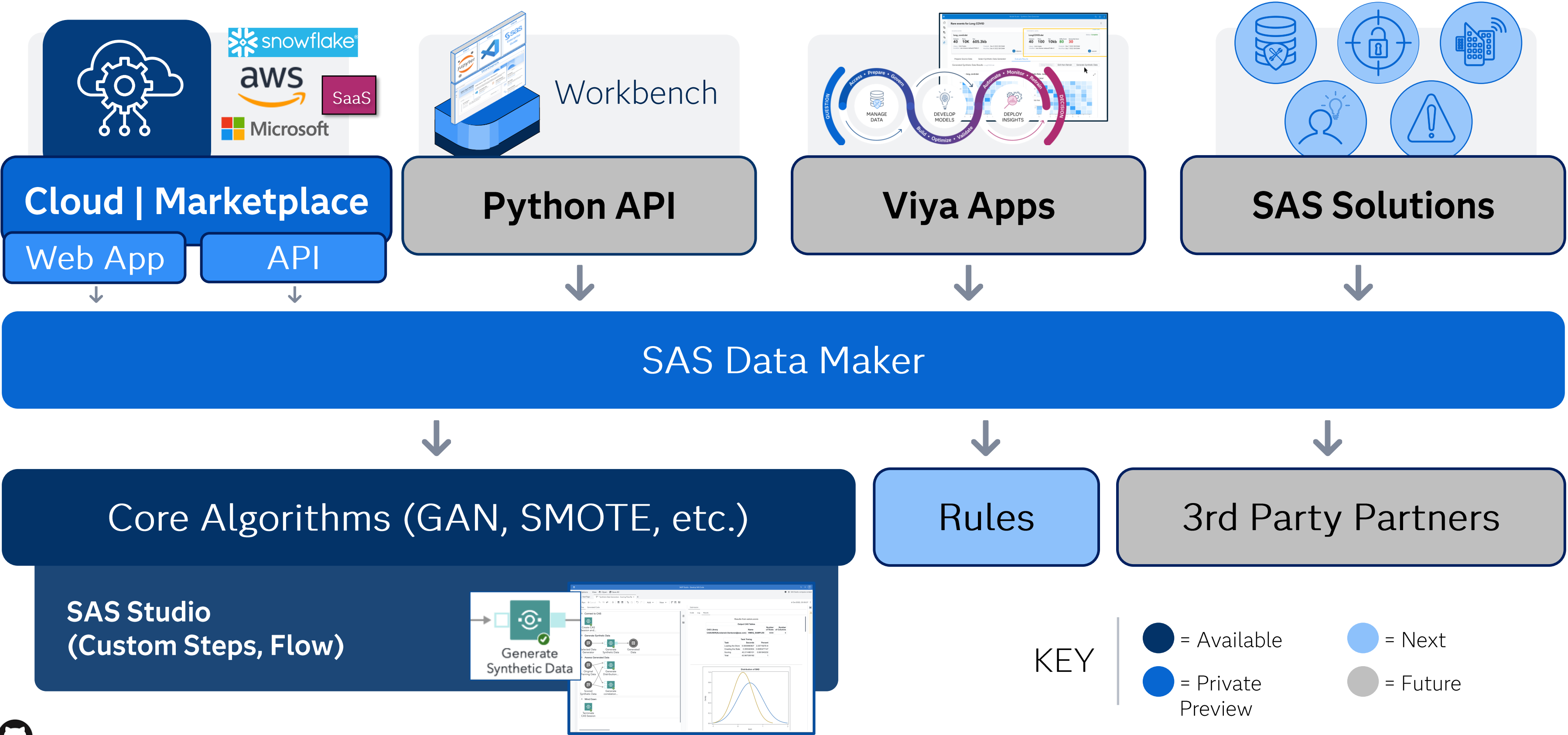


Provide Feedback

Tell us about your experience.

[Submit feedback](#)

SAS Data Maker Product Vision



[sassoftware/sas-studio-custom-steps/Synthetic Data Generation](https://github.com/sassoftware/sas-studio-custom-steps/Synthetic%20Data%20Generation)

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SAS acquires Hazy synthetic data software to boost generative AI portfolio

Strategic move reinforces SAS leadership in data and AI, enhancing customer innovation and data security

Cary, NC (Nov 12, 2024)

SAS, a global leader in data and AI, today announced the acquisition of the principal software assets of Hazy, a pioneer in [synthetic data](#) technology. This strategic acquisition aims to enhance SAS' robust data and AI portfolio, further equipping its customers with critical and timely synthetic data generation capabilities as their use of AI rapidly expands.

"Our acquisition of Hazy's IP represents a pivotal step in our commitment to innovation in the next generation of data management and AI," said Jim Goodnight, CEO of SAS. "Hazy is a pioneer in bringing synthetic data to market as a viable enterprise product, and analysts rank it among the top software providers in its category. By integrating their technology, we can offer our customers unparalleled opportunities to harness data safely and effectively, enabling them to experiment and model scenarios that were previously out of reach and gain a competitive advantage."

This move positions SAS at the forefront of data innovation, enabling more robust and secure AI applications, with future integration opportunities with SAS® Viya®. By integrating Hazy's synthetic data capabilities, SAS will empower customers to innovate and conduct deep research, overcoming challenges related to data availability, access or quality.

Leading the charge in synthetic data innovation

According to Kathy Lange, Research Director, AI Software at IDC, "Synthetic data is a game-changer for companies

https://www.sas.com/en_us/news/press-releases/2024/november/hazy-syntheticdata.html

Editorial contacts:

- **SAS Corporate HQ**
Shannon Heath (919) 531-2676

Why Synthetic Data Is Essential for Your Organization's AI-Driven Future

What is synthetic data and why do you need it?

Organizations use synthetic data generated by AI when real data is unavailable, leading to more accurate insights.

- Searchable private information
- Prohibitive cost
- Model training inefficiency
- Bias or imbalance
- Train sparse data scenarios

By 2026,
75% of businesses will use generative AI to create synthetic customer data, up from less than 5% in 2023.

Source: Gartner, "Generative AI: The Future of Synthetic AI Capabilities," 2023.

Benefits of synthetic data for organizations

60%	56%	45%	31%	30%
Improved model accuracy	Improved model efficiency	Reduced data privacy concerns	Improved AI explainability/interpretability	Reduced impact of biases

Source: SAS, "Synthetic Data: The Future of AI," 2023.

Research shows mixed feelings about the use of synthetic data

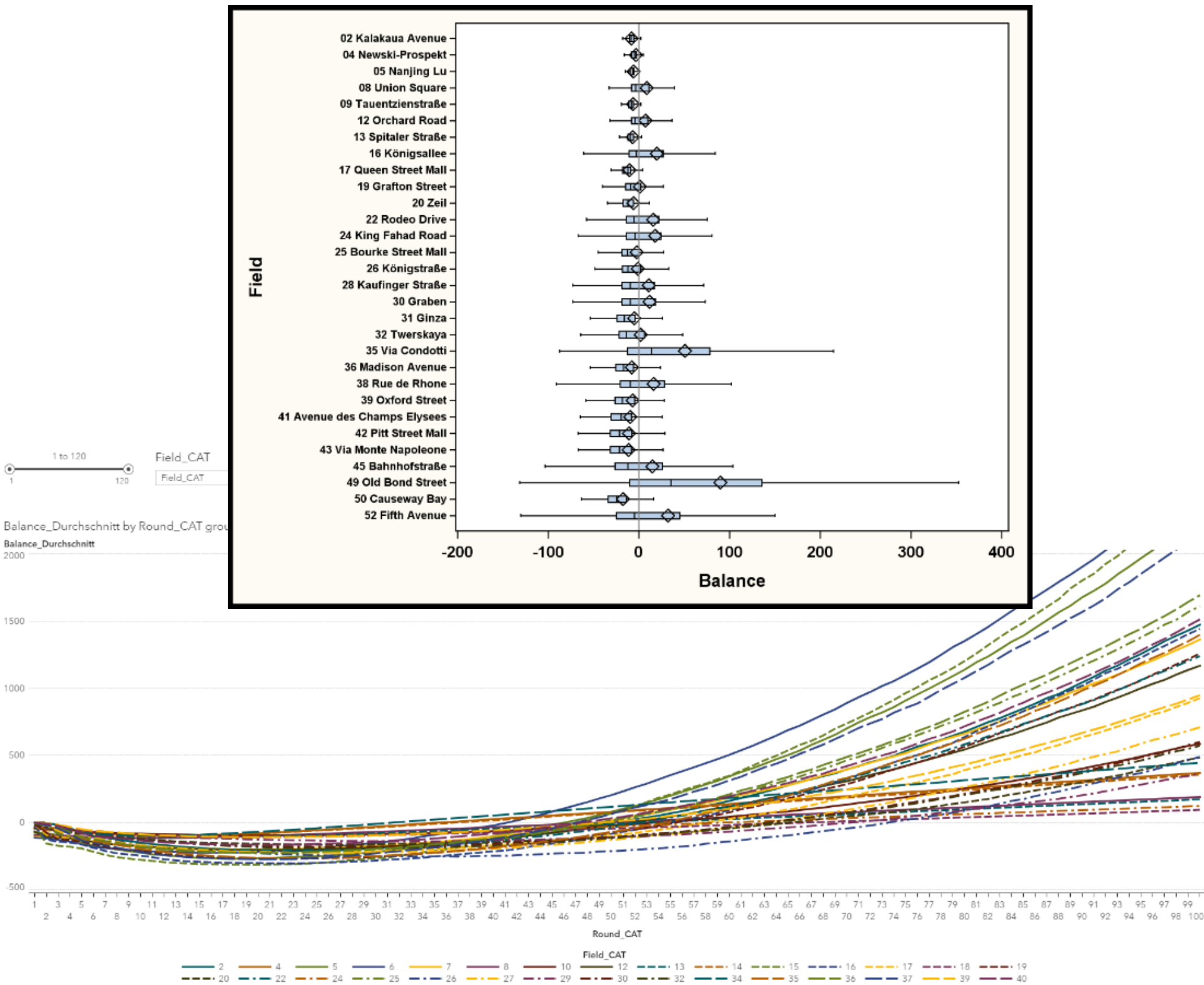
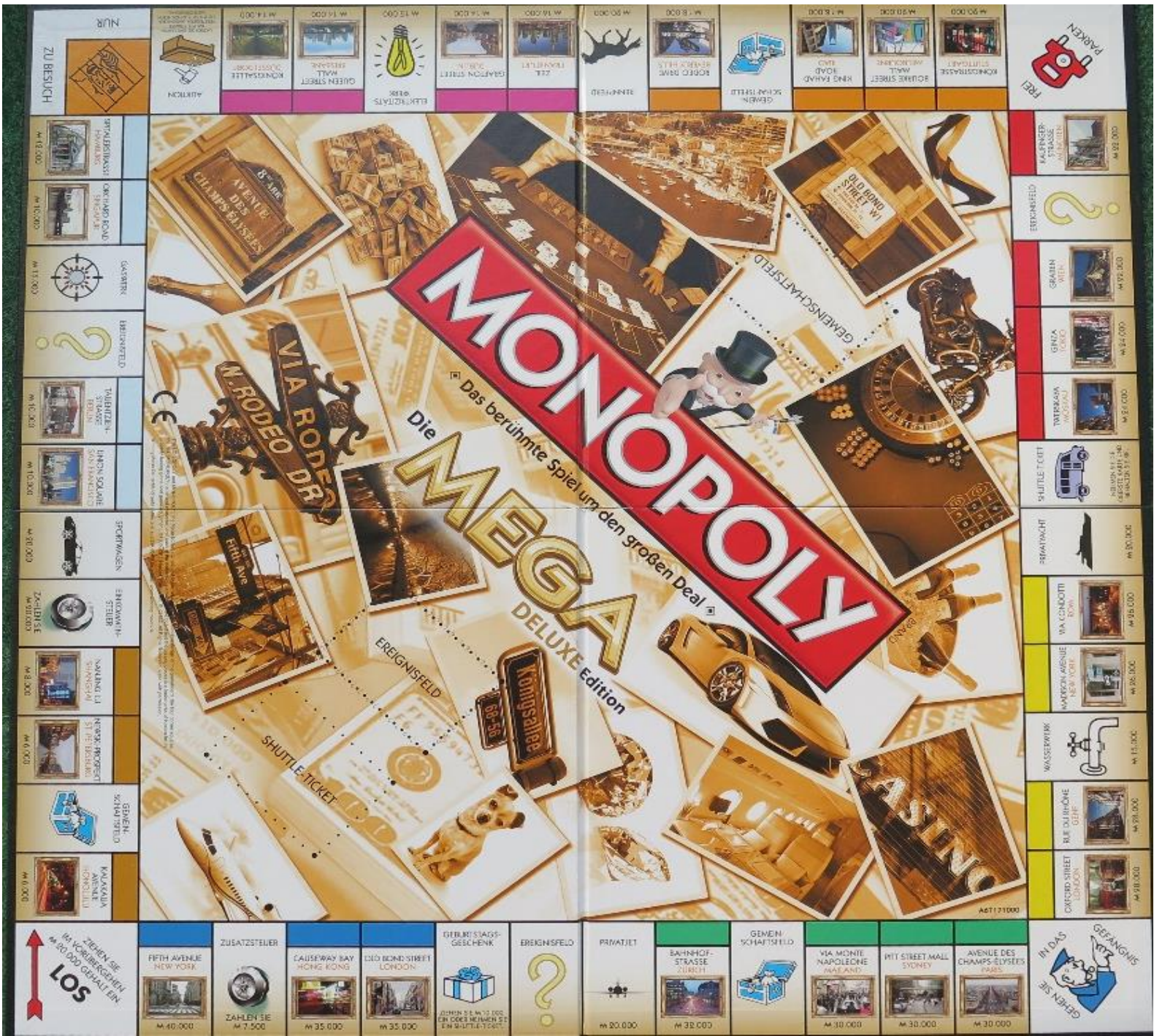
75%	80%
Organizations believe synthetic data is essential for AI innovation	Organizations believe synthetic data is essential for AI innovation

Source: SAS, "Synthetic Data: The Future of AI," 2023.

Methods for generating synthetic data

Synthetic Data from a Digital Twin

Data Monopoly Brettspiel (DKT Brettspiel)



3 Fragen, die man sich stellen könnte

1. Ist ja alles zufällig.
Daher werden alle Felder in etwa gleich häufig besucht werden.
2. Je hochpreisiger, umso besser.
Oder ist da noch mehr, was man berücksichtigen muss?
3. Warum gewinnen meine Kinder viel häufiger als ich?

Wie findet man die Antworten:

- 1000 Mal selbst Monopoly spielen
- die Positionen der Spielfigur am Spielfeld dokumentieren
- Gewinn- und Verlustrechnung pro Runde, pro Feld und pro Spieler händisch mitschreiben ???

Oder einen „Digitalen Zwilling“ programmieren

Der die synthetische Daten generiert, indem er die Spiele simuliert

Digitaler Zwilling für das Monopoly Brettspiel in SAS Viya programmiert

Declare and Initialize

Loop over Scenarios (Games)

Initialize Scenario

Loop over Rounds and Players

Generate Random Numbers

Follow Instructions

Generate Deterministic/Random Behaviour

Update Counts, Values, States

Output the Record

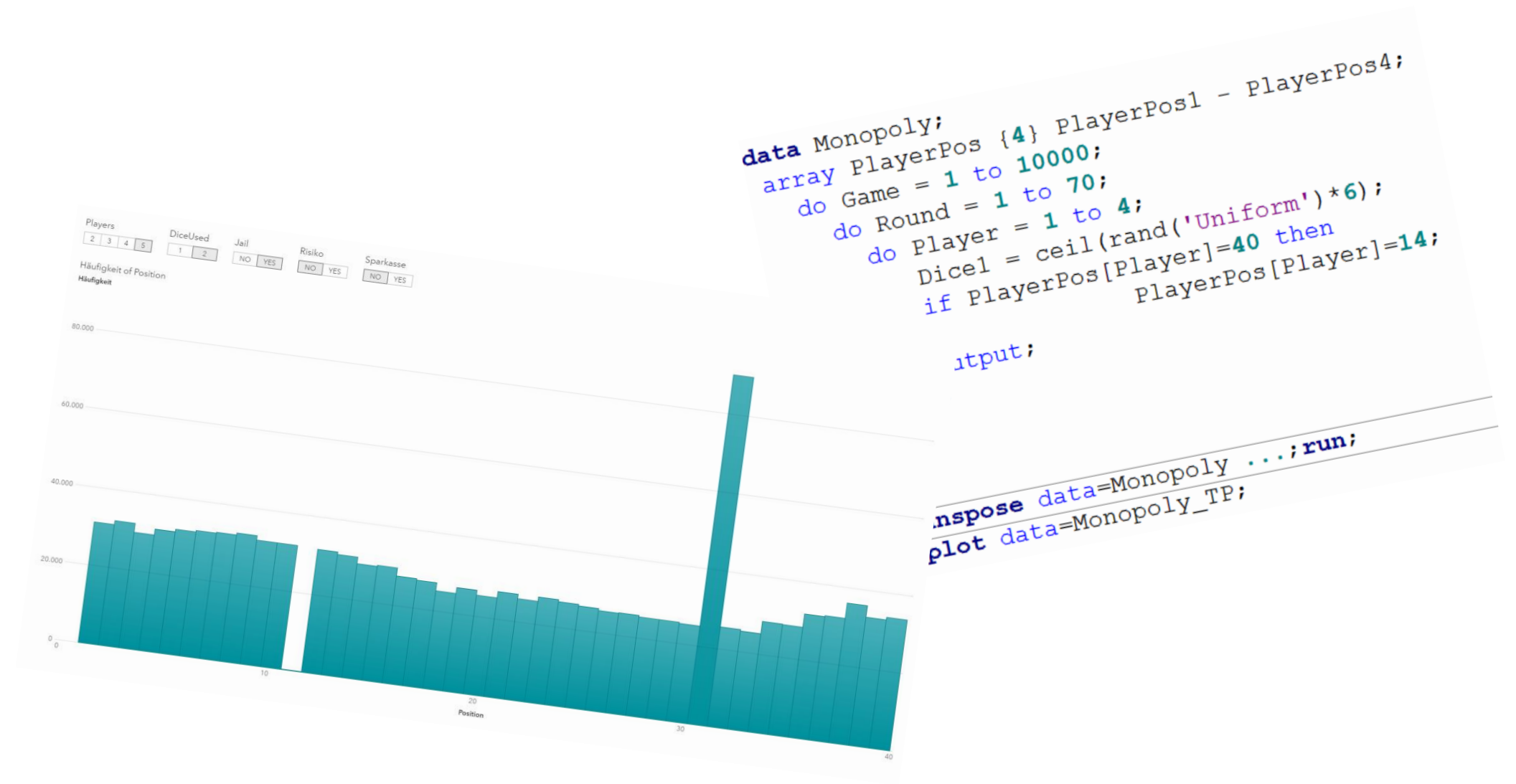
End Loop

End Loop

Prepare Analysis Data: Aggreg., Transpose, Enrich

Calculate Output Statistics, Display Output

- 10000 Spiele in weniger als einer Minute gespielt, mitdokumentiert und ausgewertet



Beispiel: Mitprotokollieren der Einnahmen/Ausgaben/Saldo

Output 28.1: Balance of Player 1 and 2 for the first 20 Rounds

Round	Pos1	Pos2	Pos3	Pos4	Balance1	Balance2	Income1	Income2	Expense1	Expense2
1	8	9	6	6	-10	-10	0	0	10	10
2	15	18	11	16	-10	-10	0	0	10	10
3	27	28	17	22	-10	-32	0	0	10	32
4	35	38	30	31	-36	-60	0	0	36	60
5	44	49	38	14	-36	-92.6	0	2.4	36	95
6	46	9	47	25	-36	-97.6	0	2.4	36	100
7	2	18	7	36	-42	-97.6	0	2.4	42	100
8	12	28	15	44	-52	-112.6	0	2.4	52	115
9	20	39	25	52	-68	-140.6	0	2.4	68	143
10	28	48	35	16	-74.8	-131.6	2.2	11.4	77	143
11	41	6	7	24	-104.8	-131.6	2.2	11.4	107	143
12	45	13	19	33	-136.8	-143.6	2.2	11.4	139	155
13	49	27	23	43	-140.3	-140.1	2.2	14.9	142.5	155
14	9	37	28	52	-143.3	-128.1	2.2	26.9	145.5	155
15	19	50	42	9	-144.3	-160.1	2.2	29.9	146.5	190
16	33	8	1	18	-143.7	-160.7	2.8	29.9	146.5	190.6
17	37	18	26	32	-143.7	-160.7	2.8	29.9	146.5	190.6
18	44	21	31	43	-143.7	-160.7	2.8	29.9	146.5	190.6
19	2	29	34	49	-148.7	-157.2	2.8	33.4	151.5	190.6
20	10	43	45	4	-145.9	-170.2	5.6	33.4	151.5	203.6

Output 28.2: Cost and Revenue of Field 28

Round	Pos1	Pos2	Pos3	Pos4	Field28	Setup28	Revenue28	Cost28	Balance28
1	8	9	6	6	0	0	0	0	0
2	15	18	11	16	0	0	0	0	0
3	27	28	17	22	2	0	0	22	-22
4	35	38	30	31	2	0	0	22	-22
5	44	49	38	14	2	0	0	22	-22
6	46	9	47	25	2	0	0	22	-22
7	2	18	7	36	2	0	0	22	-22
8	12	28	15	44	2	1	0	37	-37
9	20	39	25	52	2	1	0	37	-37
10	28	48	35	16	2	1	9	37	-28
11	41	6	7	24	2	1	9	37	-28
12	45	13	19	33	2	1	9	37	-28
13	49	27	23	43	2	1	9	37	-28
14	9	37	28	52	2	1	18	37	-19
15	19	50	42	9	2	1	18	37	-19
16	33	8	1	18	2	1	18	37	-19
17	37	18	26	32	2	1	18	37	-19
18	44	21	31	43	2	1	18	37	-19
19	2	29	34	49	2	1	18	37	-19
20	10	43	45	4	2	1	18	37	-19

Digital twin to generate synthetic sail races to train a decision support model for race tactics

```

/*****
*** PART II - START SIMULATION LOOP
*****/

*** Loop until epsilon around upwind mark has been reached;
do until ( DirectLine < &UpwindZone. or count > 200);

    count=count+1;
    racetime = racetime + &timestep.;

    if racetime <= &maxtime. then Winddir = input(RaceTime+WindSelectTime,wind_lkp.);
    else Winddir=180;

/*****
*** PART III - Analyse the Angle and decide about a FORCED tacking because of Layli
*****/

*** If AngleTack has already been set --> do nothing;
if AngleTack=1 and abs(AngleToMark) > &AngleTackLimit. then tack=0;
else do;
    *** If Layline exceeded --> TACK;
    if abs(AngleToMark) > &AngleTackLimit. then do;
        tack=1;
        AngleTack=1;
    end;
    *** otherwise continue;
else do;
    AngleTack=0;

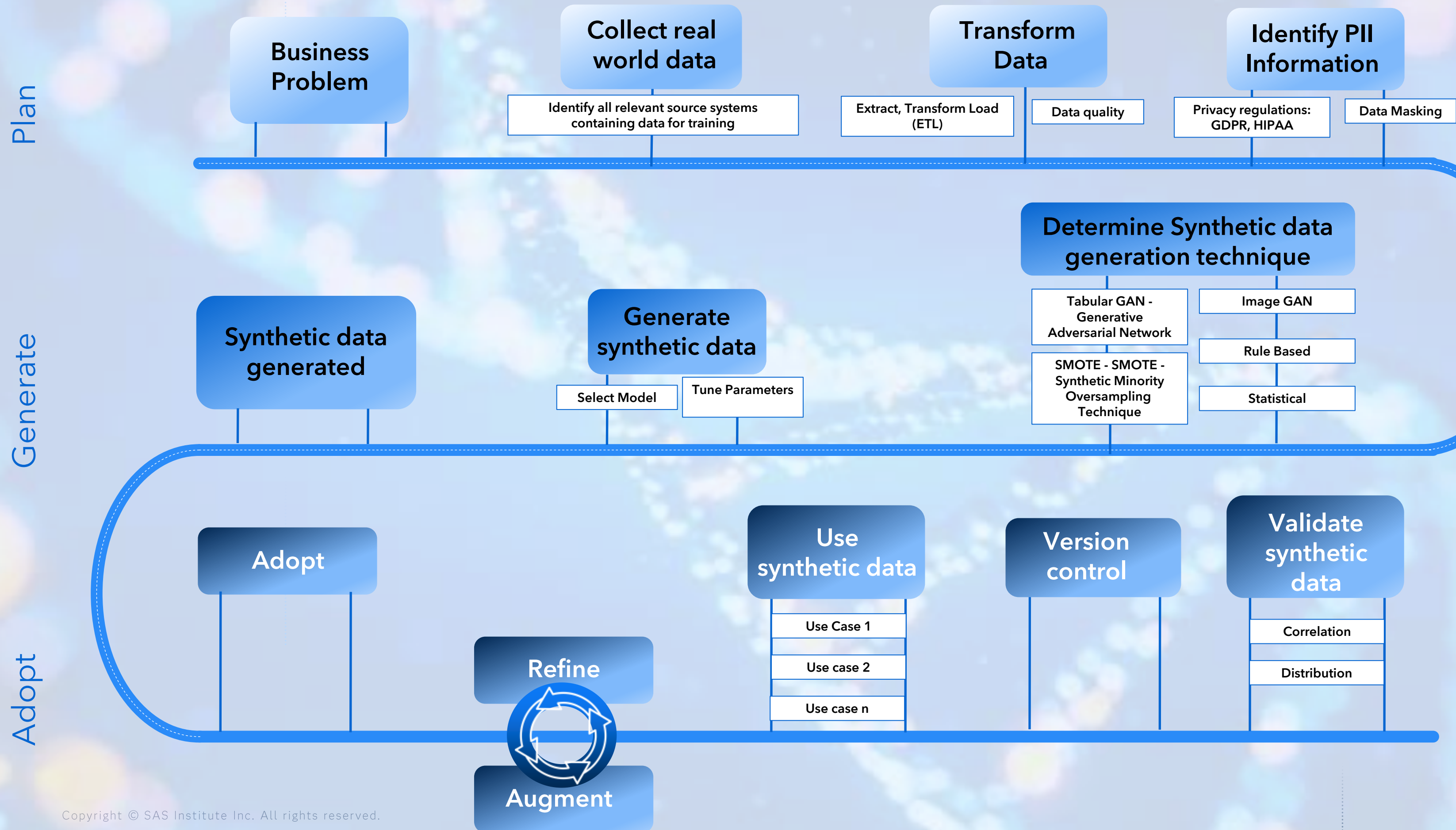
/*****
*** PART IV - INDIVIDUAL BOAT TACTICS SCORING
*****/

```

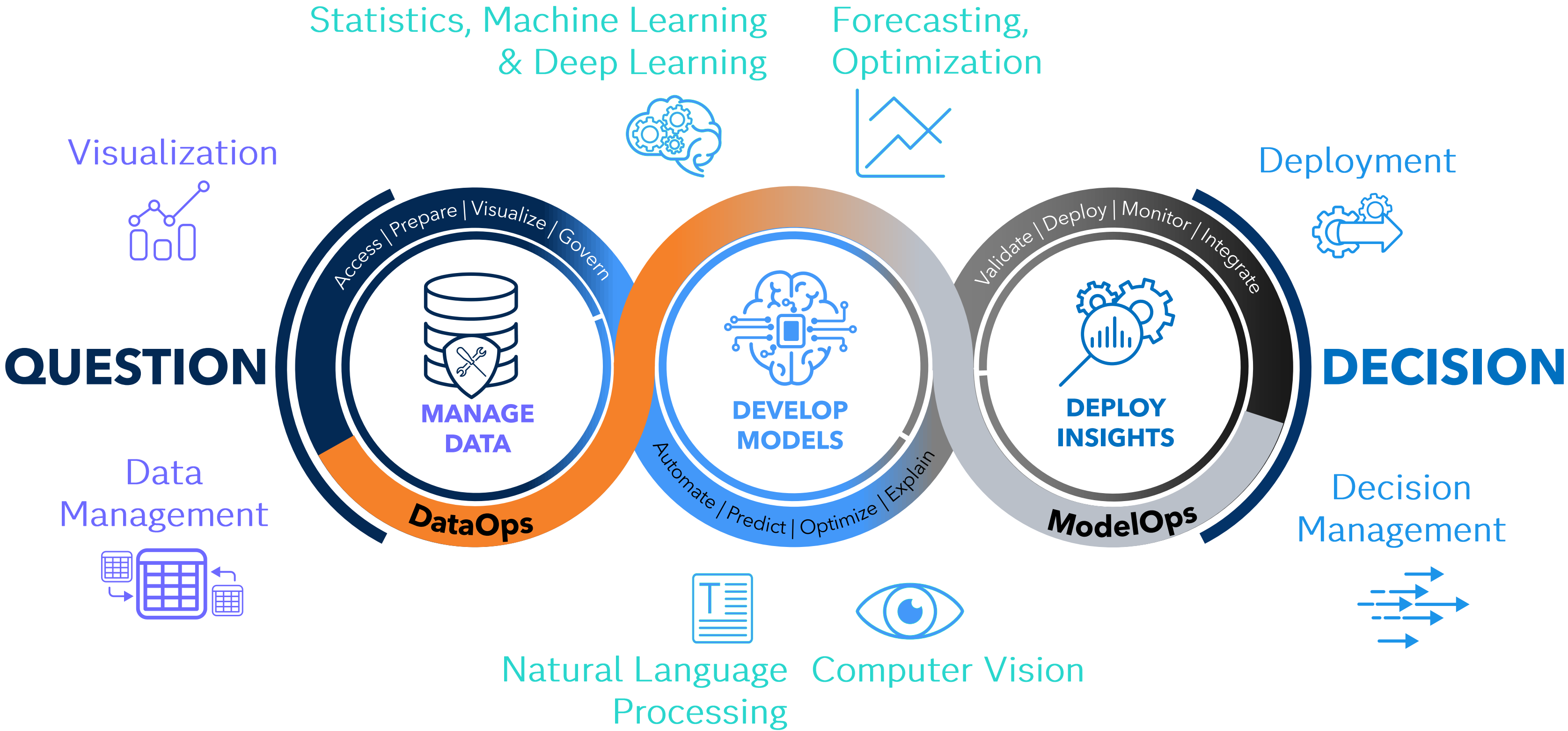


Wie sieht der Prozess der Generierung und Validierung der synthetischen Daten aus

User journey to generate Synthetic Data



From the business question to the decision



Conclusion

- Synthetic Data can help in many applications
- *Sammelst du noch oder generierst Du schon??* – Business Questions/Purpose, Data Quality+Preparation, Considerations in statistical modeling, Validation are still present!!
- SAS offers the SAS Viya Platform + SAS Data Maker for synthetic data generation
- Other offerings are e.g. <https://www.syntho.ai/>

Additional Information

Synthetic Data blogs

- [Generating Synthetic Data using Generative Adversarial Networks](#)
- [SAS for Synthetic Data Generation in Martech](#)

Accelerators for SAS Viya

- [SAS Studio Custom Steps for Synthetic Data Generation](#)

SAS Action sets for Synthetic Data Generation

- [SMOTE](#)
- [GANs](#)

Links

- Articles at SAS Communities
 - <https://communities.sas.com/t5/SAS-Communities-Library/Data-Science-and-Data-Preparation-Article-Overview-by-Gerhard/ta-p/727875>
- Books
 - [Applying Data Science - Business Case Studies Using SAS](#)
 - [Programms at github: https://github.com/gerhard1050/Applying-Data-Science-Using-SAS](https://github.com/gerhard1050/Applying-Data-Science-Using-SAS)
 - [Data Quality for Analytics Using SAS](#)
 - [Data Preparation for Analytics Using SAS](#)
- Digital Twin for the Monopoly Board Game
 - [Analytics mal anders: E-Werk, Bahnhof oder Schlossallee? Der Monopoly-Check | SAS Forum Digital 2021](#)
 - [Studying Complex Systems – Simulating the Monopoly Board Game](#)

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